LESSON 5.3

Determine the Area of Trapezoids

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

WHY THIS MATTERS Your architecture firm is designing a trapezoidal window for a modern building. The glass supplier charges by the square foot, so the team must calculate the exact area. The window has a top edge of 4 feet, a bottom edge of 8 feet, and a height of 5 feet.

REMEMBER

Determine the Area of Trapezoids

Area of Trapezoids. Formula: Area = ½ × (b1 + b2) × h A trapezoid is a four-sided shape with one pair of parallel sides. The two parallel sides are called the bases, and they have different lengths.

1. A trapezoid is a four-sided shape with one pair of parallel bases.
2. Add the two bases together: (b1 + b2).
3. Multiply by the perpendicular height, then take half

Words: **Trapezoid *(Trapecio)*** **Parallel *(Paralelas)*** **Base 1 (b1) *(Base 1 (b1))*** **Base 2 (b2) *(Base 2 (b2))*** **Height *(Altura)***

1 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 7 m and height 4 m?

(A) 11 square m
(B) 14 square m
(C) 14 m
(D) 28 square m

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 12 cm and height 5 cm?

(A) 17 square cm
(B) 30 square cm
(C) 30 cm
(D) 60 square cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A triangle has an area of 36 square inches and a base of 9 inches. What is its height?

(A) 4 inches
(B) 8 inches
(C) 27 inches
(D) 324 inches

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Why does the triangle area formula use ½?

(A) Triangles always have half as many sides
(B) The height is always half the base
(C) Area formulas always use ½
(D) A triangle is half of a parallelogram with the same base and height

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A builder is pouring concrete for a trapezoidal driveway. The driveway has parallel edges of 20 feet and 12 feet with a height of 15 feet. One bag of concrete covers 6 square feet. What is the area of the trapezoidal driveway?

(A) 47 sq ft
(B) 120 sq ft
(C) 240 sq ft
(D) 480 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 Circle the letter of the best answer. Show how you know.

A builder is pouring concrete for a trapezoidal driveway. The driveway has parallel edges of 20 feet and 12 feet with a height of 15 feet. One bag of concrete covers 6 square feet. How many bags of concrete does the builder need?

(A) 6
(B) 24
(C) 40
(D) 240

SHOW YOUR WORK

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

SENTENCE STARTER The area of the trapezoid is 1/2 x (___ + ___) x ___ = 1/2 x ___ x ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Each blueprint window lists its bases and height. Find each area, then sort the windows by whether they need more than 40 sq ft of glass.

Groups: More than 40 sq ft of glass | 40 sq ft of glass or less

_____ Window: b1=8, b2=6, h=5

_____ Window: b1=10, b2=8, h=6

_____ Window: b1=4, b2=2, h=9

_____ Window: b1=11, b2=9, h=5

_____ Window: b1=7, b2=3, h=6

_____ Window: b1=13, b2=7, h=8

2 **Write the letter of the matching item on each line.**

Match each blueprint window to the area of glass it needs.

_____ Trapezoid window b1=6, b2=4, h=3

_____ Trapezoid window b1=9, b2=5, h=4

_____ Trapezoid window b1=10, b2=6, h=5

_____ Trapezoid window b1=12, b2=8, h=6

(A) 15 sq ft

(B) 28 sq ft

(C) 40 sq ft

(D) 60 sq ft

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A builder is pouring concrete for a trapezoidal driveway. The driveway has parallel edges of 20 feet and 12 feet with a height of 15 feet. One bag of concrete covers 6 square feet. Why add the two parallel edges and then take half?

(A) Because trapezoids are half of rectangles

(B) Because the average of the two bases acts like one rectangle's width

(C) Because 20 + 12 = 32 is easier

(D) Because the height must be halved

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Blueprint 1: A trapezoidal lobby window has bases of 6 ft and 10 ft and a height of 4 ft. What is its area for the glass order?

(A) 16 sq ft

(B) 32 sq ft

(C) 40 sq ft

(D) 60 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

Blueprint 2: A trapezoidal skylight has bases of 9 ft and 7 ft and a height of 6 ft. What is its area?

(A) 44 sq ft
(B) 48 sq ft
(C) 63 sq ft
(D) 96 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 **Circle the letter of the best answer. Show how you know.**

Two equal trapezoid windows join to form a parallelogram with area 96 sq ft. What is the area of ONE trapezoid?

(A) 16 sq ft
(B) 48 sq ft
(C) 96 sq ft
(D) 192 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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